SAURABH RANJAN

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Ph.D. Cognitive Neuroscientist | AI / ML Researcher | Cognitive and Machine Psychology | NeuroAI

EDUCATION

Ph.D., Psychology — Behavioral & Cognitive Neuroscience University of Florida, USA | 2025

Dissertation: Reality Monitoring in Humans and Artificial Intelligence

M.Sc., Cognitive Science University of Allahabad, India | 2018

Thesis: Intentional Binding in Future-Directed Intentions

B.Sc., Physics Birla Institute of Technology, Mesra, India | 2016

CORE TECHNICAL STACK

AI / ML & LLMs: PyTorch, scikit-learn, LangChain, LangGraph, Pydantic, HuggingFace, prompt engineering, retrieval-augmented generation (RAG), agentic workflows, evaluation pipelines for LLM safety

***Biomedical & Health-Outcomes Methods*:** predictive modeling on biobank-scale data (UK Biobank RS-fMRI → phenotype prediction), neuroimaging analytics (MEG/EEG with MNE, fMRI with Nilearn/SPM), linearizing encoding models, temporal generalization analysis, clinical-grade reproducible pipelines

Statistical Modeling: Bayesian hierarchical models (PyMC3, brms), multinomial processing trees, network analysis (bootnet, qgraph), generalized mixed-effects models (lme4), causal & metacognitive modeling

Data Engineering & MLOps: Python (expert), R (advanced), MATLAB, Bash; pandas, NumPy, Git/GitHub, SLURM/HPC, containerized environments, large-scale behavioral data collection (Prolific, Qualtrics, PsychoPy/JS)

Visualization & Communication: seaborn, matplotlib, plotly, ggplot2, tidyverse — publication-ready figures, dashboards, and stakeholder reporting

RESEARCH EXPERIENCE

Researcher — AI for Biomedical Health Outcomes *2026*

University of Florida, Gainesville, FL

- Developing AI agents with planning, reflection, tool use, skills, and multi-agent systems for healthcare and medical decision making.
- Building reproducible analytic pipelines (Python, PyTorch, LangChain/LangGraph, SLURM/HPC) for agentic decision-making tasks.
- Collaborating on the development of Master's-level courses on Data Science and Agentic LLMs for UF AI for Biomedical Health Sciences (AIBHS).

Graduate Researcher *2020 – 2025*

Odegaard Lab & UF Department of Psychology, Gainesville, FL

- **Reality Monitoring & AI Safety:** Discovered generalizable behaviors that let humans and LLMs distinguish internally generated from externally generated information, called as reality monitoring, a core capability that fails during **hallucination in LLMs and as well as in humans**.

- Designed and ran behavioral experiments with 100+ in-person human participants and matched LLM simulations across conversation-history and feedback conditions, producing standardized human–LLM benchmarking pipelines.

- Identified a dissociation between single-trial and conversation-memory regimes in LLMs that inverts source-attribution accuracy, a finding with direct implications for how clinical assistants should manage chart history and cited evidence.

- **Neuroimaging & Predictive Modeling:** Engineered ML pipelines for MEG temporal generalization and fMRI linearizing encoding models; predicted phenotype from RS-fMRI on UK Biobank-scale data.

- **World Models as Psychological Imagination Network Analysis:** Built imagination networks across Florida, London, and Poland cohorts; used psychological network analysis to compare population-level cognitive structure between humans and synthesized LLM populations — methodology applicable to subgroup discovery in health data.

- **Open-Source Tooling:** Authored Psych Scanner, a production-grade Python framework (LangChain + LangGraph + Pydantic) for running cognitive experiments with LLMs at scale; supports persona management, conversation-history manipulation, and population-level synthesis.

- Mentored 7 undergraduate researchers; co-authored conference posters (1 award-winning); developed and taught a full segment of Physiological Psychology.

Research Assistant *2019 – 2020*

Center of Behavioral and Cognitive Sciences, University of Allahabad, India

- Performed hpothesis, design, analysis on global–local perception using array of patches and human-agency studies; contributed to an OSF preprint on future directed intentional binding.

Research Assistant *2018*

Homi Bhabha Centre for Science Education, TIFR, Mumbai, India

- Investigated in embodied cognition and vector education in

PAPERS

Working Papers

[2] **Ranjan, S.**, & Odegaard, B. Generalization of generation effect in reality monitoring.

Software

[2] **Ranjan, S.**, Makwana, M., Sokratous, K., & Odegaard, B. metasignal: A Python package for comprehensive metacognitive analysis and decision-making. Can be viewed on Preprint on Google Drive.

[1] **Ranjan, S.**, Sokratous, K., & Makwana, M.Psych Scanner: A Framework for Systematic Cognitive Evaluation of Large Language Models.

Theory and Empirical Peer-Reviewed & Preprints

[10] [Preprint] **Ranjan, S.,** Sokratous, K., & Odegaard, B. (2026). Reality Monitoring in Large Language Models: Self-Knowledge That Transforms with Conversation Memory. arXiv:2607.23927. https://arxiv.org/abs/2607.23927

[9] [Preprint] **Ranjan, S.**, & Odegaard, B. (2025). Psychological Imagination Networks Show Cross-Population Centrality and Clustering Alignment in Humans That Large Language Models Fail to Replicate. arXiv:2510.04391. https://arxiv.org/abs/2510.04391

[8] **Ranjan, S.**, & Odegaard, B. (2024). Heterarchy or hierarchy? Insights from a new model of visual imagination. *Physics of Life Reviews*, 49, 74–76.

[7] **Ranjan, S.**, & Odegaard, B. (2024). Reality monitoring and metacognitive judgments in a false-memory paradigm. *Neuroscience Research*, 201, 3–17.

[6] Maynes, R., Faulkner, R., Callahan, G., Mims, C. E., **Ranjan, S.**, Stalzer, J., & Odegaard, B. (2023). Metacognitive awareness in the sound-induced flash illusion. *Philosophical Transactions of the Royal Society B*, 378(1886), 20220347.

[5] Chiasson, P., Boylan, M. R., Elhamiasl, M., Pruitt, J. M., **Ranjan, S.**, Riels, K., ... & Keil, A. (2023). Effects of neurofeedback training on performance in laboratory tasks: A systematic review. *International Journal of Psychophysiology*.

[4] Aggarwal, A., & **Ranjan, S.** (2022). How do undergraduate students reason about ethical and algorithmic decision-making? *Proc. 53rd ACM Technical Symposium on Computer Science Education*, 488–494.

[3] Karnam, D., Agrawal, H., Parte, P., **Ranjan, S.**, Borar, P., Kurup, P. P., ... & Chandrasekharan, S. (2021). Touchy feely vectors: A compensatory design approach to support model-based reasoning. *Journal of Computer Assisted Learning*, 37(2), 446–474.

[2] Karnam, D., Agrawal, H., Parte, P., **Ranjan, S.**, Sule, A., & Chandrasekharan, S. (2019). Touchy feely affordances of digital technology for embodied interactions can enhance 'epistemic access'. *IEEE T4E*: Technology for Education, 114–121.

[1] [Preprint] **Ranjan, S.**, & Srinivasan, N. (2019). Sense of agency for future-directed intentions. doi.org/10.31234/osf.io/qa93k.

INVITED TALKS

[2] Ranjan, S. (May, 2026). *The Spark of Artificial Neuroscience: Psychologically-Grounded Evaluation of Large Language Models.* Virtual invited talk, Autonomous Empirical Research Group + Laboratory for Automated Scientific Discovery of Mind and Brain (PI: Dr. Sebastian Musslick), Osnabrück University.

[1] Ranjan, S. (March, 2026). *Controlling Imagery Generation and its Awareness.* Virtual invited talk, Robert Reinhart Lab, Boston University.

SELECTED CONFERENCE POSTERS & TALKS

[8] **Ranjan, S.**, & Odegaard, B. (2025). Psychological Imagination Networks in Humans and LLMs. Frontiers in NeuroAI, Kempner Institute Symposium, Harvard University.

[7] **Ranjan, S.**, & Odegaard, B. (2025). Visual Imagination Networks in Humans and LLMs. Vision Sciences Society Annual Meeting, St. Pete, FL.

[6] **Ranjan, S.**, & Odegaard, B. (2024). The Fragility of Reality Monitoring under Extraneous Factors. Psychonomic Society 65th Annual Meeting, NYC.

[5] **Ranjan, S.**, & Odegaard, B. (2024). Reality Monitoring, Fast and Slow [Poster + Flash Talk]. Association for Psychological Science, San Francisco. *Awarded Scott O. Lilienfeld APS Travel Award.*

[4] Maw, M., Zhuang, L., Baltes, J., **Ranjan, S.**, & Odegaard, B. (2024). Task demands and sensory externalization in reality monitoring. PGSO Undergraduate Research Forum, UF. *Mentee won 3rd Prize.*

[3] Dundigalla, S., Johnson, D., Roh, A., **Ranjan, S.**, & Odegaard, B. (2024). Cognitive strategies in reality monitoring. PGSO Undergraduate Research Forum, UF.

[2] **Ranjan, S.**, Baltes, J., Roh, A., & Odegaard, B. (2023). Confidence in reality monitoring judgments. *Journal of Vision*, 23(9), 4844.

[1] **Ranjan, S.**, & Srinivasan, N. (2018). Intentional binding and future-directed intentions [Talk]. Annual Conference of Cognitive Science, IIT-Guwahati, India.

AWARDS & FELLOWSHIPS

- 2025 — College of Liberal Arts and Sciences Travel Award, University of Florida

- 2025 — Threadgill Dissertation Fellowship, University of Florida
- 2024 — Scott O. Lilienfeld APS Travel Award, Association for Psychological Science
- 2024–2025 — UF Department of Psychology Travel Awards (×4)
- 2020–2025 — Graduate Student Fellowship, UF Department of Psychology
- 2016–2018 — Graduate Merit Scholarship, Centre of Behavioral & Cognitive Sciences, University of Allahabad